

ENGINEERING MECHANICS (EM)

UNIT – 1

1. A ladder 6m long and with 300N weight is resting against a wall at an angle of 60 to the ground. A man weighing 750N climbs the ladder. At what position along the ladder from bottom does he induce slipping? The coefficient of friction for both wall and the ground with ladder is 0.2.
2. Derive a relation for the efficiency of a screw jack, taking friction into account
3. An open belt running over two pulleys 1200 mm and 500 mm diameters connects two parallel shafts 4000 mm apart. The maximum tension in the belt is 1830 N. Coefficient of friction between the belt and the pulley is 0.3. The driven pulley of 1200 mm diameter runs at 200 rpm. Determine :a) Power transmitted b) torque on each of two shafts
4. Derive an expression for length of crossed belt drive.
5. What is angle of repose? Prove that angle of repose is equal to the angle of friction.
6. A shaft running at 100 rpm drives another shaft at 200 rpm and transmits 12kW. The belt is 100 mm wide and 12mm thick and coefficient of friction is 0.25. the distance between the shaft is 2.5 m and the diameter of smaller pulley is 500 mm. Calculate the stress in (i) an open belt and (ii) crossed belt, connecting two pulleys

UNIT – 2

1. Define Lami's theorem?
2. A machine weighing 5KN is supported by two chains attached to some points on the machine. One chain goes to hook in the ceiling and has an inclination of 45 degrees with horizontal. The other chain goes to eye bolt in the wall and is inclined at 30 degrees to the horizontal. Find tensions induced in the chains
3. Derive an equation for resultant using parallelogram law of forces
4. Find the magnitude of the two forces such that if they act at right angles their resultant is $\sqrt{10}$ N. But if they act at 60 degrees their resultant is $\sqrt{13}$ N
5. Explain the different types supports and support reactions
6. State and explain the theorem of Varignon?

UNIT – 3

1. Find the centroid of a semi circular lamina of radius R from its base?

2. State and prove parallel axis theorem of area moment of inertia.
3. State and prove perpendicular axis theorem.
4. Determine the moment of Inertia of a triangle with respect to its Centroidal Xaxis parallel to the base?
5. Derive an expression for the moment of inertia of a circular ring of uniform cross section of radius 'R' about its diametrical axis
6. Derive an expression for the moment of inertia of a circular plate of radius 'R' and thickness 't' about its centroidal axis.

UNIT – 4

1. Define about polar moment of inertia?
2. Explain about transfer theorem?
3. Derive mass moment of inertia?
4. Derive moment of inertia of masses?
5. Derive transfer formula for mass moments of inertia?
6. Derive mass moments of inertia for composite bodies?

UNIT – 5

1. The motion of a particle in rectilinear motion is defined by the relation where s is in meters and t in seconds. Find (a) Acceleration of particle when the velocity is zero (b) the position and the total distance travelled when acceleration is zero.
2. An auto is accelerated from rest to a speed of 100kmph and then immediately decelerated to a stop. If the total elapsed time is 20sec, determine the distance covered. The acceleration and deceleration are both constant but not necessarily of the same magnitude
3. A train travelling 96kmph has to slow down an amount of work being done on the line. Instead of continuing a constant speed it, therefore moves with a constant retardation of 1.6kmph/s until the speed is reduced of 24kmph. It is then travels at a constant speed for 400m and then accelerates at 0.8kmph/s until its speed is once more 90kmph. Find the delay period
4. The distance covered by a freely falling body in the last one second of its motion and that covered in the last but one second are in the ratio 5:4. Calculate the height from which the body was dropped and the velocity with which it strikes the ground
5. A stone is thrown vertically upwards with a velocity of 19.6m/s from the top of a tower 24.5m height. Calculate the a) time required for the stone to reach the ground. b) velocity of the stone in its downward travel at the point in the same level as the point of projection. c) The maximum height to which the stone will rise in its flight.

UNIT – 6

1. Derive an expression for kinetic energy of a body rotating about a fixed axis
2. A bullet of mass 25gms moving horizontal with a velocity of 600m/s strikes a wooden block of mass 5Kg resting on a rough horizontal surface. The bullet after striking the block remains buried in the block and both travels a distance of 90cms before coming to rest. Determine: (a) average resistance between the block and horizontal surface (b) coefficient of friction between block and horizontal surface
3. Prove that the work done by body is equal to change in its kinetic energy
4. A hammer of mass 1500Kg drops from a height of 60cms on a pile of mass 750Kg. find (i) the common velocity after impact assuming plastic impact (ii) average resistance of the ground.
5. Define the law of conservation of momentum and prove it.
6. What is a simple pendulum? Derive an equation for the time period?